

Load Interconnection Request System Impact Study Report

For Queue Project Q21-1789

Chatom 138 kV
July 27, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1789. The Interconnection Customer proposes a 20.8 MW large load (data center) facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on June 21, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 406 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$618,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Salamancarocheater St - Declo 138 KV line
Contingency: Aspen - Crandall 345 KV
Loading: 102.0% (Rating: 296 MVA)
DFAX: 9.8%
Pre-project loading: 101.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Salamancarocheater St - Declo 138 KV line	Reconductor line	\$19,990,000	3% (\$618,000)
	Total allocated to project		\$618,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.